Supplementary Material

Threat-Model Dependence in the Adversarial Robustness of Mamba-Based Network Intrusion Detection

Haydar Kilic
Department of Artificial Intelligence Engineering, Ostim Technical University, Ankara, Turkey
haydar.kilic@ostimteknik.edu.tr

This document holds the material the main paper refers to as released with the archived artefact. Section S1 gives the derivations and definitions omitted from the paper's methodology section and Section S2 the two pseudo-code listings; Section S3 extends the statement of scope; Sections S4–S9 give in full the analyses the paper reports only in summary. Every quantity is reproduced from the result files deposited with the artefact, and the file each comes from is named where it is used. Section S10 indexes the deposited tables and figures.

The main paper cites the sections of this document as S1–S10. Code, result frames and this document are archived together at `https://doi.org/10.5281/zenodo.23497561` (concept identifier; release v2.0.1); the corpus subsamples are in the companion record `https://doi.org/10.5281/zenodo.23499680`. This document is not self-contained and is to be read with the paper.

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S1 Derivations and definitions omitted from the paper

S1.1 Class-weighted empirical risk

Operational NIDS traffic is dominated by benign records, and an unweighted empirical risk admits a minimiser that predicts the majority class unconditionally. The detector parameters are therefore estimated by minimising the class-weighted risk

$$\boldsymbol{\theta}^{\star} = \arg\min_{\boldsymbol{\theta}} \frac{1}{|\mathscr{D}|} \sum_{i=1}^{|\mathscr{D}|} w_{y^{(i)}}\, \ell\big(f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}), y^{(i)}\big), \tag{S1}$$

with ℓ the categorical cross-entropy and the weights the inverse-frequency coefficients

$$w_k = \frac{|\mathscr{D}|}{K\, n_k}, \qquad n_k = \sum_{i=1}^{|\mathscr{D}|} \mathbb{1}\big[y^{(i)} = k\big], \quad K = 2. \tag{S2}$$

Equation (S2) is the exact analogue of the balanced class-weight convention adopted for the Random Forest baseline, so the imbalance correction is identical across the deep and non-deep arms of the comparison. The predictive distribution and decision rule are

$$p_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}) = \operatorname{softmax}\big(f_{\boldsymbol{\theta}}(\boldsymbol{x})\big), \qquad \hat{y}(\boldsymbol{x}) = \arg\max_{k \in \{0,1\}} f_{\boldsymbol{\theta}}(\boldsymbol{x})_k. \tag{S3}$$

S1.2 Standardisation and the projection operator

Continuous features are standardised with statistics estimated exclusively on the training partition and applied unchanged to the test partition,

$$\tilde{x}_j = \frac{x_j - \mu_j^{\mathrm{tr}}}{\sigma_j^{\mathrm{tr}} + \varepsilon}, \qquad \varepsilon = 10^{-8}, \tag{S4}$$

and the Euclidean projection onto the feasible set of the paper's Equation (1) is applied component-wise,

$$\Pi_{\mathscr{S}_\varepsilon(\boldsymbol{x})}(\boldsymbol{z}) = \operatorname{clip}\Big(\boldsymbol{x} + \operatorname{clip}(\boldsymbol{z} - \boldsymbol{x},\, -\varepsilon,\, \varepsilon),\; l,\; u\Big), \tag{S5}$$

where $l = \min_{\mathscr{D}_{\mathrm{tr}}} \boldsymbol{x}$ and $u = \max_{\mathscr{D}_{\mathrm{tr}}} \boldsymbol{x}$ are the empirical bounds of the training distribution in standardised feature space.

S1.3 Continuous-time formulation and zero-order-hold discretisation

A linear time-invariant state-space model maps a scalar input $u(t)$ to an output $y(t)$ through a latent state $\boldsymbol{h}(t) \in \mathbb{R}^N$:

$$\frac{\mathrm{d}\boldsymbol{h}(t)}{\mathrm{d}t} = \boldsymbol{A}\boldsymbol{h}(t) + \boldsymbol{B}u(t), \qquad y(t) = \boldsymbol{C}\boldsymbol{h}(t) + D\,u(t), \tag{S6}$$

with $\boldsymbol{A} \in \mathbb{R}^{N\times N}$, $\boldsymbol{B} \in \mathbb{R}^{N\times 1}$, $\boldsymbol{C} \in \mathbb{R}^{1\times N}$ and a skip coefficient $D \in \mathbb{R}$. Discretising (S6) with a zero-order hold over a step size $\Delta > 0$ yields

$$\boldsymbol{h}_t = \overline{\boldsymbol{A}}\boldsymbol{h}_{t-1} + \overline{\boldsymbol{B}}u_t, \tag{S7}$$

$$y_t = \boldsymbol{C}\boldsymbol{h}_t + D\,u_t, \tag{S8}$$

where, following the reference formulation of Gu and Dao with a diagonal $\boldsymbol{A}$,

$$\overline{\boldsymbol{A}} = \exp(\Delta \boldsymbol{A}), \qquad \overline{\boldsymbol{B}} = \Delta \boldsymbol{B}. \tag{S9}$$

The expression for $\overline{\boldsymbol{B}}$ is the first-order term of the exact zero-order-hold coefficient $(\Delta\boldsymbol{A})^{-1}\big(\exp(\Delta\boldsymbol{A}) - \boldsymbol{I}\big)\Delta\boldsymbol{B}$, whose expansion is $\Delta\boldsymbol{B} + O(\Delta^2)$; the truncation is adopted for numerical stability and for consistency with the reference implementation. The state matrix is parameterised as

$$\boldsymbol{A} = -\exp(\boldsymbol{A}_{\log}), \qquad \boldsymbol{A}_{\log} \in \mathbb{R}^{d_{\text{inner}} \times N}, \tag{S10}$$

so that every entry of $\boldsymbol{A}$ is strictly negative and, since $\Delta > 0$, every entry of $\overline{\boldsymbol{A}}$ lies in $(0,1)$: the recurrence is contractive, which precludes the exploding-state pathology of naive recurrent parameterisations. Under the S4D-real initialisation the n-th diagonal entry is set to $\boldsymbol{A}_{\log} = \log n$ for $n = 1, \ldots, N$, giving $\boldsymbol{A} \in \{-1, \ldots, -N\}$; this is the sequence the source writes as $A_n = -(n+1)$ under zero-based indexing. With $N = 16$ the resulting decay rates span a sixteenfold range, so the state channels integrate the input over time-scales differing by more than an order of magnitude.

S1.4 The selective projection and the block dimensions

The selective parameters are obtained by a single projection of the internal representation $\boldsymbol{u}_t \in \mathbb{R}^{d_{\text{inner}}}$:

$$\big[\tilde{\boldsymbol{\delta}}_t, \boldsymbol{B}_t, \boldsymbol{C}_t\big] = \boldsymbol{W}_x \boldsymbol{u}_t, \tag{S11}$$

whose three blocks have widths d_{inner}, N and N respectively, so that $\tilde{\boldsymbol{\delta}}_t \in \mathbb{R}^{d_{\text{inner}}}$, $\boldsymbol{B}_t, \boldsymbol{C}_t \in \mathbb{R}^N$ and $\boldsymbol{W}_x \in \mathbb{R}^{(d_{\text{inner}}+2N)\times d_{\text{inner}}}$, with $\boldsymbol{W}_\Delta \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{inner}}}$. In the block equations of the paper the affine maps $\boldsymbol{W}_{\text{in}} \in \mathbb{R}^{2d_{\text{inner}} \times d_{\text{model}}}$ and $\boldsymbol{W}_{\text{out}}$ are applied row-wise, that is, independently at each step, whereas the depth-wise causal convolution and the selective scan act along the sequence axis; the gating endows the block with a multiplicative input-dependent pathway and the residual is post-normalised, the normalisation being applied after the residual is added.

S1.5 Detector head

The detector embeds the univariate pseudo-sequence, adds a learned positional embedding $\boldsymbol{P} \in \mathbb{R}^{L \times d_{\text{model}}}$ initialised from a truncated normal distribution with $\sigma = 0.02$, applies $K = 3$ stacked blocks, mean-pools over the sequence axis and classifies:

$$\boldsymbol{h}^{(0)} = \boldsymbol{x}\boldsymbol{W}_e + \boldsymbol{P}, \qquad \boldsymbol{W}_e \in \mathbb{R}^{1 \times d_{\text{model}}}, \tag{S12}$$

$$\boldsymbol{h}^{(k)} = \text{S6Block}_k\big(\boldsymbol{h}^{(k-1)}\big), \quad k = 1, \ldots, K, \tag{S13}$$

$$\bar{\boldsymbol{h}} = \frac{1}{L}\sum_{t=1}^{L} \boldsymbol{h}_t^{(K)}, \tag{S14}$$

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{W}_2\, \text{GELU}\big(\boldsymbol{W}_1\, \text{LayerNorm}(\bar{\boldsymbol{h}})\big), \tag{S15}$$

with $\boldsymbol{W}_1 \in \mathbb{R}^{(d_{\text{model}}/2)\times d_{\text{model}}}$ and $\boldsymbol{W}_2 \in \mathbb{R}^{2\times(d_{\text{model}}/2)}$.

S1.6 Closed-form evaluation used in the second validation check

For diagonal $\boldsymbol{A}$ the time-varying recurrence admits the closed form

$$\boldsymbol{h}_t = \sum_{s=1}^{t} \Big(\prod_{r=s+1}^{t} \overline{\boldsymbol{A}}_r\Big) \odot \overline{\boldsymbol{B}}_s \otimes \boldsymbol{u}_s, \tag{S16}$$

in which the empty product at $s = t$ is the identity. Because $\overline{\boldsymbol{A}}_r = \exp(\Delta_r \otimes \boldsymbol{A})$, the product in (S16) is evaluated without a recurrence as $\exp(\boldsymbol{S}_t - \boldsymbol{S}_s)$, where $\boldsymbol{S}_t = \sum_{r \le t} \Delta_r \otimes \boldsymbol{A}$ is a cumulative sum in log space; since $\boldsymbol{A} \le 0$ and $\Delta_r > 0$, every such factor lies in $(0,1]$ and the evaluation is numerically stable. This is the reference against which the sequential loop is compared in the paper's second validation check, which reports a maximum deviation of 1.39×10^{-17} in double precision (`results/verification/verification.json`, field `check2` $= 1.3877787807814457 \times 10^{-17}$).

S1.7 Optimisation schedule

The learning rate is annealed by a cosine schedule stepped once per epoch,

$$\eta_e = \tfrac{1}{2}\eta_0\Big(1+\cos\big(\pi e/N_{\mathrm{ep}}\big)\Big), \qquad N_{\mathrm{ep}} = 40, \tag{S17}$$

the epoch budget being written N_{ep} to avoid collision with the expansion factor E.

Model selection and early stopping are governed by the balanced accuracy on the validation split, which the paper states in words and which is

$$\mathrm{BA} = \frac{1}{K}\sum_{k=1}^{K}\frac{\sum_i \mathbb{1}[\hat{y}^{(i)}=k]\,\mathbb{1}[y^{(i)}=k]}{\sum_i \mathbb{1}[y^{(i)}=k]}, \qquad K = 2, \tag{S18}$$

the unweighted mean of the per-class recalls. It is used in place of raw accuracy because a degenerate majority-class predictor attains a raw accuracy equal to the majority prior while attaining $\mathrm{BA} = 0.5$, and because it is the quantity the class weights of Equation (S2) make the training objective an analogue of, so selection and optimisation target the same criterion.

S1.8 Per-category detection rates

For every attack category k with at least five test instances the clean and adversarial detection rates are

$$D_k^{\mathrm{clean}} = \frac{1}{|I_k|}\sum_{i\in I_k}\mathbb{1}\big[\hat{y}(\boldsymbol{x}_i)=1\big], \tag{S19}$$

$$D_k^{\mathrm{adv}} = \frac{1}{|I_k|}\sum_{i\in I_k}\mathbb{1}\big[\hat{y}(\boldsymbol{x}_i^{\mathrm{adv}})=1\big], \tag{S20}$$

where $I_k = \{i : y_i = 1,\, c_i = k\}$, and the evasion success rate of the paper's Equation (16) is their difference. The analogous quantities for the two binary classes are computed by the same procedure and are released with the artefact rather than tabulated, since the category-level decomposition subsumes them for the malicious class.

S1.9 Latency and throughput

Per-sample latency and throughput follow from the measured per-batch wall-clock time $\bar{\lambda}_{\mathrm{batch}}$ in milliseconds at batch size b as

$$\lambda_{\mathrm{sample}} = \frac{\bar{\lambda}_{\mathrm{batch}}}{b}, \qquad \mathrm{TP} = \frac{1000\,b}{\bar{\lambda}_{\mathrm{batch}}}. \tag{S21}$$

Measurement uses CUDA events with explicit device synchronisation after 20 warm-up iterations over 100 timed repetitions. The Random Forest baseline executes on CPU and is timed with a host-side clock over a reduced schedule of five warm-up and twenty timed iterations, so its figures are reported for reference only.

S1.10 Paired testing and the Holm–Bonferroni correction

Because the seed is shared between the proposed model and its comparator, the comparison is paired, and the parametric statistic is

$$t = \frac{\bar{d}}{s_d/\sqrt{S}}, \qquad d_s = a_s^{\mathrm{Mamba}} - a_s^{\mathrm{comp}}, \tag{S22}$$

over $S = 7$ seeds. Since one hypothesis is tested per (dataset, attack) cell, the family-wise error rate over the resulting family of twelve is controlled by the Holm–Bonferroni step-down procedure: with ordered raw p-values $p_{(1)} \le \cdots \le p_{(m)}$, the adjusted values are

$$\tilde{p}_{(j)} = \max_{i\le j}\ \min\big\{1,\ (m-i+1)\,p_{(i)}\big\}, \tag{S23}$$

and $H_{(j)}$ is rejected iff $\tilde{p}_{(j)} \le \alpha = 0.05$. The two conventions the paper describes differ only in the choice of comparator; the adjusted values therefore differ between them even where the raw p-value is identical, because the correction is applied within each family separately. The clearest instance is NF-UNSW-NB15 under transfer, where the raw p-value of 0.0223 becomes $\tilde{p} = 0.1558$ in the pre-specified family and $\tilde{p} = 0.1113$ in the post-hoc family (`results/tables/T2_istatistiksel_anlamlilik.csv` and `T2b_istatistiksel_anlamlilik_posthoc.csv`). Every significance claim in the paper is made with respect to the pre-specified family.

S2 Pseudo-code

Algorithm 1: Leakage-aware partitioning. The strongest guarantee supported by the metadata present in a given corpus is selected: a group split when flow or endpoint identifiers exist, a chronological split when a parsable timestamp exists, and a stratified random split only as a last resort, in which case a leakage caveat is written to the run log.

Input: Records $\mathscr{D}$; candidate group columns G; timestamp column τ; test fraction $\rho_{\text{te}} = 0.2$; fixed split seed $s = 42$, shared by all experimental seeds
Output: Partition assignment $\pi \in \{\text{train}, \text{test}\}^N$

$G^* \leftarrow$ columns of G present in $\mathscr{D}$;
if $G^* \neq \emptyset$ **then**
  $k \leftarrow$ composite key from the first two columns of G^*;
  if $|\text{unique}(k)| \ge 10$ **then**
    $\pi \leftarrow \textsc{GroupShuffleSplit}(\mathscr{D}, k, \rho_{\text{te}}, s)$;
    return π ; `// no group spans both partitions`
if τ *present* ***and*** $\ge 90\%$ *of* τ *parses* **then**
  sort indices by τ (stable);
  assign the latest $\lceil \rho_{\text{te}} N \rceil$ records to *test*;
  return π ; `// chronological hold-out`
$\pi \leftarrow \textsc{StratifiedSplit}(\mathscr{D}, y, \rho_{\text{te}}, s)$;
record an explicit leakage caveat in the run log;
return π;

Algorithm 2: Selective scan (S6) forward pass for one block. B is the batch size, L the sequence length, N the state dimension and $d_{\text{inner}} = E\, d_{\text{model}}$ the expanded channel width. $\odot$ is the Hadamard product and $\otimes$ broadcasts along the state dimension. Lines 7–10 realise the time-varying recurrence as an explicit loop.

Input: $\boldsymbol{x} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$; parameters $\boldsymbol{W}_{\text{in}}, \boldsymbol{W}_x, \boldsymbol{W}_\Delta, \boldsymbol{W}_{\text{out}}, \boldsymbol{A}_{\log}, \boldsymbol{D}$, the depth-wise convolution kernel $\boldsymbol{K} \in \mathbb{R}^{d_{\text{inner}} \times k}$ and the LayerNorm scale and shift
Output: out $\in \mathbb{R}^{B \times L \times d_{\text{model}}}$

$[\boldsymbol{u}', \boldsymbol{z}] \leftarrow \boldsymbol{W}_{\text{in}} \boldsymbol{x}$;
$\boldsymbol{u} \leftarrow \text{SiLU}\big(\text{DWConv}_{k=4}(\boldsymbol{u}')[:, :L]\big)$; `// causal`
$[\tilde{\boldsymbol{\delta}}, \boldsymbol{B}, \boldsymbol{C}] \leftarrow \boldsymbol{W}_x \boldsymbol{u}$;
$\Delta \leftarrow \text{softplus}(\boldsymbol{W}_\Delta \tilde{\boldsymbol{\delta}})$; $\boldsymbol{A} \leftarrow -\exp(\boldsymbol{A}_{\log})$;
$\overline{\boldsymbol{A}} \leftarrow \exp(\Delta \otimes \boldsymbol{A})$; $\overline{\boldsymbol{B}} \leftarrow \Delta \otimes \boldsymbol{B}$;
$\boldsymbol{h} \leftarrow \boldsymbol{0} \in \mathbb{R}^{B \times d_{\text{inner}} \times N}$;
for $t \leftarrow 1$ **to** L **do**
  $\boldsymbol{h} \leftarrow \overline{\boldsymbol{A}}_{:,t} \odot \boldsymbol{h} + \overline{\boldsymbol{B}}_{:,t} \otimes \boldsymbol{u}_{:,t}$;
  $\boldsymbol{y}_t \leftarrow \sum_{n=1}^{N} \boldsymbol{h}_{:,:,n} \boldsymbol{C}_{:,t,n}$;
$\boldsymbol{y} \leftarrow \text{stack}(\boldsymbol{y}_1, \ldots, \boldsymbol{y}_L) + \boldsymbol{D} \odot \boldsymbol{u}$;
$\boldsymbol{y} \leftarrow \boldsymbol{y} \odot \text{SiLU}(\boldsymbol{z})$;
return $\text{LayerNorm}(\boldsymbol{W}_{\text{out}} \boldsymbol{y} + \boldsymbol{x})$;

S3 Extended statement of scope

The complete experimental grid comprises three datasets, five architectures, seven random seeds and ten attack settings (FGSM and PGD at $\varepsilon \in \{0.01, 0.03, 0.05, 0.1\}$ and black-box transfer at $\varepsilon \in \{0.03, 0.1\}$). Because the Random Forest is excluded from all ten by construction, the master frame holds 945 evaluation records rather than the 1,155 of a full product: 105 clean cells over five architectures and 840 attacked cells over the four differentiable ones.

The official NSL-KDD files were re-counted directly to confirm the protocol statement the paper makes about them: `KDDTrain+` holds 125,973 records carrying 23 distinct labels and `KDDTest+` holds 22,544 records carrying 38, of which 17 labels occur in the test file and in no training record. Those 17 are `apache2`, `httptunnel`, `mailbomb`, `mscan`, `named`, `processtable`, `ps`, `saint`, `sendmail`, `snmpgetattack`, `snmpguess`, `sqlattack`, `udpstorm`, `worm`, `xlock`, `xsnoop` and `xterm`, and they account for 3,750 test records, 16.63% of the partition. The difference $38 - 23$ does not give 17 because two training labels (`spy` and `warezclient`) are absent from the test file; the count of unseen types is therefore a set difference and not a subtraction of cardinalities.

After sanitisation and tensorisation the three corpora comprise 125,973/22,544 (NSL-KDD, official split), 111,975/27,289 (NF-UNSW-NB15) and 250,143/62,370 (CSE-CIC-IDS2018) training/test records, with malicious proportions of 0.465/0.569, 0.503/0.490 and 0.529/0.500. The record counts written by the split verification (`results/verification/verification.json`) precede the sanitisation stage, which removes the non-finite rows: for CSE-CIC-IDS2018 that file reports 250,918/62,730, which sanitisation reduces to the 250,143/62,370 above.

The native feature dimensionalities are 41, 41 and 79, so the fixed pseudo-sequence length $L = 64$ entails zero-padding of the trailing 23 coordinates, 35.94% of the sequence, for the first two corpora and the truncation of 15 trailing attributes for CSE-CIC-IDS2018. The automated health audit reported no residual NaN/Inf entries in any partition and identified 24, 23 and 8 near-zero-variance features respectively. These counts are largely an artefact of the pseudo-sequence construction rather than of the underlying telemetry: for NF-UNSW-NB15 all 23 are padded positions, and for NSL-KDD 23 of the 24 are, the remaining one being a single genuinely constant input attribute. CSE-CIC-IDS2018 is truncated rather than padded and therefore contributes no padded positions at all, so all 8 of the columns flagged on that corpus are attributes that are genuinely constant across its training partition.

Counting a category as analysable when at least five instances survive in the test partition retained four categories for NSL-KDD (DoS, Probe, R2L, U2R), nine for NF-UNSW-NB15 and two for CSE-CIC-IDS2018 (Bot and Infiltration, the latter spelled `Infilteration` in the released label column). The last of these three counts is a consequence of the partitioning rather than of the corpus: the file carries fourteen distinct attack labels, but the chronological cut distributes them almost disjointly, twelve occurring only within the training interval, Infiltration alone being present on both sides of the cut, and Bot occurring exclusively in the test partition. Detection on that corpus is thus partly a zero-shot task.

Neither the group-split check nor the chronological-split check can be reproduced from the released tensors, since the address and timestamp columns are dropped before tensorisation; both are therefore performed from the raw corpora and their output is released in `results/verification/`. The group split on `IPV4_SRC_ADDR` and `IPV4_DST_ADDR` resolved to 242 unique host pairs with zero groups shared between the two partitions, and the timestamp column of CSE-CIC-IDS2018 parsed for every record (`parsed_fraction` $= 1.0$) with the training and test intervals confirmed disjoint.

S4 The NF-UNSW-NB15 leakage episode in full

S4.1 The defect and the route by which it was found

The released NF-UNSW-NB15-v2 CSV carries 45 columns, of which two are labels: the binary `Label` and `Attack`, the attack-family name. The preprocessing stage of the first execution excluded the former but not the latter, and because `Attack` is a text column it was label-encoded and admitted as a feature.

Its value set for benign flows is disjoint from its value set for malicious ones, so a membership rule on that one coordinate reproduces the label exactly, at an accuracy of 1.0000 on both partitions.

The single-threshold audit did not catch it, and the reason is exact rather than incidental. Alphabetical encoding of the ten values places `Benign` third, as code 2 behind `Analysis` and `Backdoor`, so every single cut assigns at least one class wrongly: the best of them, $x > 2$, misclassifies the 2,299 `Analysis` and 2,169 `Backdoor` flows as benign and therefore scores $1 - 4{,}468/139{,}264 = 0.9679$ on the released file, against the 1.0000 a membership test returns. Both figures are reproducible from the deposited subsample:

```
d = pd.read_csv("NF-UNSW-NB15-v2.csv")
cats = sorted(d.Attack.unique())
x = d.Attack.map({c: i for i, c in enumerate(cats)})
((x != cats.index("Benign")) == d.Label).mean()   # 1.0
((x > 2) == d.Label).mean()                       # 0.967917
```

A gap of 0.032 between the two audits is what separated a corpus that looked merely saturated from one in which a single coordinate carried the label. The defect was found instead by comparing the feature count the pipeline reported against the count the feature-importance analysis used, and the corpus was reprocessed and retrained with `Attack` withheld. Every NF-UNSW-NB15 figure in the paper comes from that corrected execution; NSL-KDD and CSE-CIC-IDS2018 were never affected and were verified record by record to be identical to their original values.

The episode is reported rather than silently repaired because the detection route matters. A membership test over per-class value sets catches this failure and the threshold test that is conventional in leakage audits does not, so an audit built only on the latter will pass a corpus in which one coordinate determines the label.

S4.2 Why saturation persists after the correction

That a 200-tree Random Forest attains 0.9976 accuracy with a seed dispersion of 2.5×10^{-5} is the pattern one expects from a dominant, near-deterministic feature rather than from a hard problem solved well. We therefore examined the impurity-based feature importances of that model together with the best accuracy obtainable from each feature under a single threshold (`results/verification/verification.json`, field `leakage`). Three features are individually almost sufficient: `MAX_TTL` and `MIN_TTL` each separate the two classes at 0.9948 accuracy with a single threshold and `MIN_IP_PKT_LEN` at 0.9916, and together they carry 0.576 of the total impurity-based importance mass.

The time-to-live fields are the more diagnostic of the two kinds. A TTL value is a function of the operating system that emitted the packet and of the number of hops it traversed, not of whether the traffic is hostile; that it predicts the label at 99.5% indicates that benign and malicious traffic in this capture originate from hosts that are distinguishable by network position or platform. Predictive power of this kind is a property of the testbed rather than of intrusion, and partitioning cannot remove it, since it is carried in the feature vector itself.

S4.3 The feature-removal ablation

Whether the models rely on these features is a separate question, and it was tested directly rather than inferred. Table S1 reports the result, computed on the group-aware split (`results/verification/ttl_ablation.json`).

The near-deterministic features are therefore removable, not load-bearing, and the natural reading that the corpus is saturated because of a small artefact is not supported. What the ablation shows instead is more general and, for the purposes of the study, more limiting. With the three features withheld, the strongest remaining single feature, `SERVER_TCP_FLAGS`, still separates the classes at 0.8573 accuracy under one threshold against a majority-class baseline of 0.5103, and each of the ten most important

Table S1. Feature-removal ablation on NF-UNSW-NB15. Held-out accuracy of the Random Forest under the group-aware split, with the near-deterministic features withheld in turn. The 41 features of the all-features row are the 43 non-label columns of the released CSV less the two endpoint-address fields reserved as group keys.

Features withheld	d	Accuracy	Change
none	41	0.997618	—
`MAX_TTL`	40	0.997655	$+3.7 \times 10^{-5}$
`MAX_TTL`, `MIN_TTL`	39	0.997508	-1.1×10^{-4}
+ `MIN_IP_PKT_LEN`	38	0.997545	-7.3×10^{-5}

remaining features exceeds that baseline by between 0.17 and 0.35. Separability is thus distributed redundantly across many attributes rather than concentrated in a few. A benchmark with this property cannot discriminate between architectures on clean performance, because any reasonable model recovers the same decision from a different subset of features, and that is precisely the pattern the clean results show: four of the five architectures separate by 1.8×10^{-3}, an interval of the same order as the largest of their seed dispersions, with CNN1D trailing them by a further 2.8×10^{-3}.

Two figures in this subsection are easily confused and are therefore distinguished explicitly. The single-threshold accuracy of `SERVER_TCP_FLAGS` is 0.8633 when all features are present and 0.8573 once the three near-deterministic features are withheld; the paper quotes the latter, because the claim it supports concerns what remains after their removal.

S5 Padding-masked re-evaluation in full

The pseudo-sequence construction pads both NSL-KDD and NF-UNSW-NB15 from 41 to 64 positions, so 35.94% of each input vector is a constant zero. An ℓ_∞-bounded adversary perturbs every coordinate, including the padding, and therefore spends roughly a third of its budget on dimensions that are constant in training. Padding coordinates have no counterpart in any emittable flow, so evasion attributable to them is an artefact of the representation rather than a security finding, which makes the size of their contribution a question about the validity of the comparison and not merely about its interpretation.

The white-box grid was therefore repeated on the two padded corpora with the perturbation confined to the informative coordinates, the padding positions being held at their clean value of zero at every step. The replication is reduced to the seeds $\{13, 42\}$ for cost and was executed on an NVIDIA A100-SXM4-40 GB device (`results/e3_padding_mask.json`). Before the masked arm is read, the harness itself was validated against the published unmasked figures on all 24 (corpus, model, quantity) checks: it reproduces the clean and FGSM values to within 8.9×10^{-5} and the PGD values, which begin from a random point in the ball, to within 3.4×10^{-3}, so the two arms are comparable.

Table S2. Padding-masked re-evaluation. Mean robust accuracy over the white-box grid with the perturbation applied to all coordinates (unmasked) and to the informative coordinates only (masked), averaged over the two corpora, both attacks and four budgets.

Model	Unmasked	Masked	Change
Transformer	0.8605	0.8648	+0.0044
Mamba	0.6633	0.7286	+0.0652
CNN1D	0.6608	0.7500	+0.0892
BiLSTM	0.5563	0.7155	+0.1592

The masked attack proves weaker for every architecture, which settles the direction of the effect: perturbing the padding assists the adversary rather than wasting its budget. The assistance is strongly architecture dependent, as Table S2 shows. The Transformer is almost indifferent to the padding, its two arms on NF-UNSW-NB15 differing by less than 2×10^{-3} under either attack, whereas the state-space detector derives a measurable part of its apparent fragility from coordinates no adversary can reach.

The dependence is also unevenly distributed in a way a narrower experiment would have misread. Under FGSM on NSL-KDD alone the state-space delta is $+0.0070$, which taken by itself would suggest padding independence; the dependence is concentrated in PGD, where it reaches $+0.1005$ on NF-UNSW-NB15 and $+0.0998$ on NSL-KDD.

The consequence for the principal finding is that its direction survives and its magnitude does not. On this subset the mean white-box margin in favour of the Transformer falls from $+0.1971$ unmasked to $+0.1363$ masked, a reduction of 31%, and the architecture ordering is unchanged in both arms. No cell inverts: the margin stays positive in all four (corpus, attack) combinations, narrowing most under PGD on NF-UNSW-NB15, from $+0.2934$ to $+0.1947$, and not at all under FGSM on NSL-KDD ($+0.0732$ to $+0.0735$). The attention detector therefore remains the more robust architecture under a realistic adversary, but by roughly two thirds of the margin the unmasked grid reports, and the larger figure should not be quoted without that qualification.

S6 What the aggregate metrics conceal about severity

Aggregate robust accuracy states how much detection is lost, not which intrusions are lost with it, and the per-category decomposition shows that the answer is architecture dependent and not stable across corpora. On NSL-KDD the volumetric categories dominate for the convolutional and recurrent baselines, with evasion rates of 0.773 and 0.804 for DoS and 0.757 and 0.472 for Probe, whereas for the state-space and attention detectors the largest rates occur on U2R (0.194) and Probe (0.167) respectively while DoS evasion remains at 0.068 and 0.073. Clean detection rates for R2L and U2R are low for every architecture (0.104–0.313), so the corresponding evasion rates are bounded from above by a small quantity.

On NF-UNSW-NB15 clean detection remains at or above 0.9840 in 35 of the 36 cells, the exception being CNN1D on Worms (0.9600), where one instance of 25 is missed, and under attack the four architectures separate cleanly: the Transformer registers at most 0.004 (DoS) across all nine categories, Mamba's largest rates are 0.143 (Generic), 0.125 (Shellcode) and 0.036 (DoS) with 0.000 on Worms and 0.001 on Reconnaissance, BiLSTM spans 0.080 (Worms) to 0.938 (Analysis), and CNN1D is evadable in every category but Worms, reaching 0.994 (Analysis), 0.982 (Shellcode) and 0.967 (Backdoor).

On CSE-CIC-IDS2018 clean detection of the malicious categories is weak in absolute terms, between 0.015 (CNN1D) and 0.149 (Mamba) for Infiltration and between 0.000 (CNN1D) and 0.006 (Transformer) for Bot, so evasion rates are correspondingly small, the largest being 0.132 (Mamba, Infiltration). One negative value is recorded, -0.494 for BiLSTM on Bot, where the adversarial detection rate (0.498) exceeds the clean rate (0.004): the perturbation moves that architecture from labelling almost every Bot flow benign to labelling about half of them malicious. A reversal of this kind is what one expects when the clean decision on a category is essentially uninformative, and Bot is precisely the family the chronological split places wholly outside the training interval.

A caveat concerns what the evasion success rate measures. Being defined on malicious instances alone, it is blind to how an architecture treats benign traffic, and the two can move in opposite directions. On NF-UNSW-NB15 under FGSM at $\varepsilon = 0.1$ Mamba retains 0.955 of its malicious detection on the first seed while its aggregate accuracy on that seed falls to 0.473, which places its accuracy on the benign class at 0.011: the perturbation drives it to label almost every benign flow malicious. CNN1D fails in the opposite direction on the same corpus, holding 0.974 on the benign class while its malicious detection falls to 0.140. A low evasion rate is therefore not by itself evidence of robustness, and the two-class decomposition released with the artefact should be read alongside the category table.

S7 Adversarial training: effective, but insufficiently verified

FGSM-mixed adversarial training raises mean white-box robust accuracy in all six (dataset, model) combinations, by between $+0.1734$ and $+0.2506$ for the state-space detector and between $+0.2005$ and $+0.2498$ for the convolutional baseline, at a clean-accuracy cost that never exceeds 0.0167 in magnitude and is of mixed sign (`results/defense_results.csv`). The gains are largest at the largest budget,

where the undefended models were weakest, and for both architectures they fall in the same cell: NF-UNSW-NB15 under PGD at $\varepsilon = 0.1$, where the state-space detector rises from 0.2508 to 0.9762 and the convolutional baseline from 0.3888 to 0.9129.

One qualification governs how far these gains can be taken, and it is the reason the paper treats the defence arm as incompletely validated. The defended checkpoints were evaluated under the white-box grid only; no black-box transfer attack was run against them. Single-step adversarial training is known to be prone to catastrophic overfitting, in which robustness to the single-step attack rises sharply while robustness to multi-step attacks collapses, and the resulting model exhibits a degenerate loss surface on which a single gradient poorly reflects the loss. Two observations argue against that pathology here: the gains are realised under PGD as well as FGSM, and they are realised at every budget rather than only at the one the defence was trained at. But the decisive check is the transfer-attack check, precisely because it does not rely on the gradient the defence may have obscured, and it was not performed. The surrogates it requires are already trained and released, so the experiment is inexpensive; until it is run, the defence is validated only by the class of attack whose gradients its training may have shaped.

S8 Threat-model realism

Perturbations in this study are generated in standardised feature space and are constrained only by the empirical feature box of the paper's Equation (1). They are not required to correspond to flows that an adversary could actually emit, and the gap between the two is not incidental. A NetFlow record's fields are linked by protocol semantics and by arithmetic identities: byte and packet counters are non-negative integers, duration and throughput are related by definition, flag fields take values from a discrete set, and a perturbation that raises the packet count without raising the byte count describes a flow that cannot exist. An unconstrained ℓ_∞ perturbation violates such relations freely.

Three consequences follow, and the paper states them rather than leaving them to be inferred. The reported robust accuracies are pessimistic as estimates of operational risk, since part of the measured degradation is caused by inputs no adversary can produce. The defence gains may be correspondingly optimistic, since a defence hardened against unconstrained perturbations has been hardened partly against a fiction. And whether the architecture ranking is stable under a constraint-aware attack is unknown: the literature is divided on whether domain constraints confer material robustness, with one line of work reporting that up to 82% of the adversarial examples PGD produces violate learned constraints and that projecting onto the constrained set restores up to 34% of accuracy, and another reporting that constraint-compliant attacks still reach misclassification rates above 95% although an adversary restricted to five randomly chosen features, about a tenth of that space, succeeds just under 50% of the time.

Establishing which reading holds for this comparison requires either a projection onto the feasible set after each attack step or an attack that searches within it. Neither was performed here, and the paper's robustness figures should be read as bounding the attainable evasion rate from above rather than as certifying an end-to-end network-level exploit.

S9 Capacity ablation in full

All results in the paper rest on a single capacity setting shared by every architecture, and the clean-accuracy comparison was therefore repeated at $d_{\text{model}} \in \{32, 64\}$ over the two seeds $\{13, 42\}$ with all remaining hyperparameters fixed (`results/ablation_results.csv`).

Two inversions must be recorded as qualifications of the principal findings. The four-way ordering is capacity-invariant only on NF-UNSW-NB15, where Mamba leads at both widths; on NSL-KDD the leading architecture changes with width, CNN1D being strongest at 32 (0.8005) and BiLSTM at 64 (0.8139), and on CSE-CIC-IDS2018 the ordering inverts outright, the two highest-scoring models at 32 being the two lowest at 64. With two seeds and a dispersion reaching ± 0.0834 in the CSE-CIC-IDS2018 cells, that inversion is most parsimoniously read as noise rather than as a capacity effect, which is itself the finding: on a corpus where models do not learn, an ablation cannot discriminate.

Table S3. Capacity ablation. Mean clean accuracy over two seeds at two embedding widths.

Dataset	Model	$d_{\text{model}} = 32$	$d_{\text{model}} = 64$	Change
NSL-KDD	Mamba	0.7435	0.7658	+0.0223
	Transformer	0.7850	0.7936	+0.0086
	CNN1D	0.8005	0.7549	−0.0456
	BiLSTM	0.7868	0.8139	+0.0271
NF-UNSW-NB15	Mamba	0.9970	0.9971	+0.0001
	Transformer	0.9966	0.9967	+0.0001
	CNN1D	0.9872	0.9949	+0.0076
	BiLSTM	0.9947	0.9964	+0.0016
CSE-CIC-IDS2018	Mamba	0.5694	0.5153	−0.0541
	Transformer	0.5705	0.5166	−0.0539
	CNN1D	0.5040	0.5460	+0.0420
	BiLSTM	0.5136	0.5681	+0.0545

The pairwise relation carrying the central comparison of the study is also not uniform. The Transformer exceeds Mamba on clean accuracy in four of the six (corpus, width) cells and trails it in the two NF-UNSW-NB15 cells by 0.0003 and 0.0004. That margin is substantial only on NSL-KDD, at 0.0415 at $d_{\text{model}} = 32$ and 0.0277 at 64, in both cases larger than either model's seed dispersion; across the four cells on the other two corpora its magnitude never exceeds 0.0013, well inside the dispersion observed on CSE-CIC-IDS2018, so those four cells establish no ordering in either direction.

Two restrictions bound what the ablation can settle in any case. It varies clean accuracy alone, and therefore offers no evidence on whether the robustness findings, which are the actual subject of the study, are artefacts of the chosen capacity; and with two widths and two seeds it lacks the power to resolve the CSE-CIC-IDS2018 inversion.

S10 Index of the deposited tables and figures

The tables and figures the paper omits are deposited with the artefact rather than reproduced here, in the forms below. Aggregated tables are in `results/tables/` as CSV, Markdown and LaTeX; the rendered copies used during preparation are in `published/tables_tex/` and `published/figures/`, with PNG versions in `published/figures_png/`.

Table S4. Deposited tables.

File	Contents
`T1_ana_ozet_sonuclari`	All five detection metrics per (dataset, model, attack, ε) cell, mean and standard deviation
`T2_istatistiksel_...`	Primary paired comparison against the pre-specified comparator
`T2b_..._posthoc`	Secondary comparison against the per-cell strongest competitor
`T3_saldiri_turu_...`	Evasion success rate per attack category
`T4_cikarim_verimliligi`	Parameter counts, latency and throughput at both batch regimes
`T5_savunma_etkinligi`	Defended against undefended, per (dataset, model, attack, ε)
`T6_hiperparametre_...`	Capacity ablation

One further frame is deposited under `results/superseded/`: the result file of the first execution of the grid, before the `Attack` leakage of Section S4 was corrected. No figure in the paper comes from it. It is kept so that one claim which would otherwise rest on testimony can be checked, namely that the obfuscated-gradient diagnostic (5) fired before the correction and does not after it: in that frame the Transformer's accuracy on NF-UNSW-NB15 never falls below 0.9976 at any budget, so a tenfold increase in ε costs it at most 0.0024, whereas in the corrected run the same tenfold increase costs 0.051

under FGSM and 0.046 under PGD. A leaking feature can therefore imitate gradient masking on this diagnostic, and the two are separated only by auditing the features rather than the curves.

Table S5. Deposited figures, in `published/figures/`.

File pattern	Contents
`clean_acc_*`	Clean accuracy by architecture, one panel per corpus
`robust_fgsm_*`	Robust accuracy against ε under FGSM, one per corpus
`robust_pgd_*`	The same under PGD
`catvuln_*`	Evasion success rate by attack category, representative cells
`latency_bs1`, `latency_bs512`	Measured latency at the two batch regimes
`params_vs_robust`	Parameter count and latency against mean robust accuracy
`defense_*`	Robustness curves before and after adversarial training
`ablation_*`	Clean accuracy against embedding width, one per corpus

The two verification records are `results/verification/verification.json`, which holds the two numeric S6 implementation checks, the split verification and the single-threshold leakage audit, and `results/verification/ttl_ablation.json`, which holds the feature-removal ablation of Section S4. The padding-masked re-evaluation of Section S5 is in `results/e3_padding_mask.{json,csv,tex}`. The resolved run configuration, including every seed, is in `results/logs/config.json`.